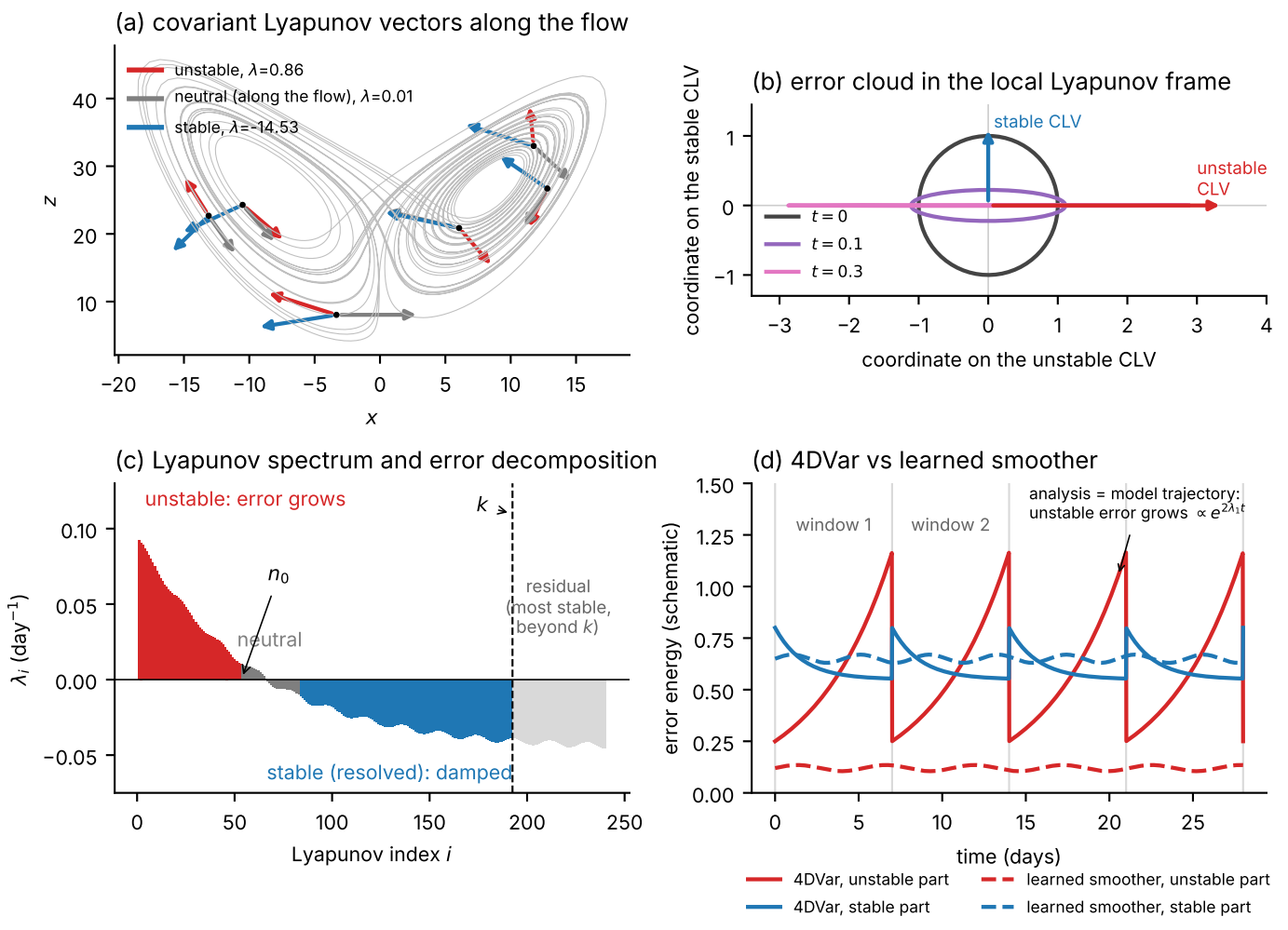

(a) covariant Lyapunov vectors along the flow
unstable, λ=0.86
neutral (along the flow), λ=0.01
stable, λ=-14.53
z
x
(b) error cloud in the local Lyapunov frame
stable CLV
unstable CLV
t = 0
t = 0.1
t = 0.3
coordinate on the stable CLV
coordinate on the unstable CLV
(c) Lyapunov spectrum and error decomposition
unstable: error grows
k
n0
neutral
residual (most stable, beyond k)
stable (resolved): damped
λi (day−1)
Lyapunov index i
(d) 4DVar vs learned smoother
window 1
window 2
analysis = model trajectory:
unstable error grows ∝ e2λ1t
error energy (schematic)
time (days)
4DVar, unstable part
4DVar, stable part
learned smoother, unstable part
learned smoother, stable part